

Reproduction and Multi-Dataset Evaluation of a Multi-Resolution Vision Transformer for Deformable Cardiac Image Registration

Ecem Karaoglu

Department of Electrical and Electronics Engineering
Middle East Technical University

July 1, 2026

Abstract

This report reproduces and evaluates DIR-MRVIT, a CNN-Transformer hybrid network for cardiac motion estimation. The aim was to take the public source code of the method, read the paper, and apply it to three public cardiac MR datasets: ACDC17, M&Ms20, and CMRxM22. The model code is used as published. The preprocessing, the cross-validation, and the evaluation scripts were written from scratch and adapted to each dataset. The reproduced method reaches Dice, surface-distance, and Jacobian scores close to the paper. The remaining gaps come from preprocessing choices, an unspecified training schedule in the paper, and random training variation. The code is publicly available at https://github.com/ecemkaraoglu/cardiac_deformable_registration.

Contents

1	Introduction	3
1.1	What This Project Is About	3
1.2	Clinical Motivation	3
1.3	What Image Registration Does	3
1.4	Project Objective	3
1.5	Use of AI Assistance	3
2	Overview of the DIR-MRVIT Method	3
2.1	Problem Formulation	4
2.2	Three-Level Pyramid	4
2.3	Global-Local Cooperative Learning	4
2.4	Convolutional Projection Transformer Block (CPTB)	4
2.5	Loss Function	5
2.6	Training Scheme	5
3	Datasets	5
4	Pipeline Overview	6
5	Implementation Details	6
5.1	Reused vs. Re-implemented Code	6
5.2	Training Setup	7
5.3	Common Preprocessing	7
5.4	Hausdorff Distance: HD95 vs Strict Maximum	7

6	Dataset-Specific Modifications	8
6.1	ACDC17	8
6.1.1	Resampling and Cropping	8
6.1.2	Stratified Five-Fold Split	8
6.1.3	Training Schedule	8
6.1.4	Surface Metrics	8
6.2	M&Ms20	8
6.2.1	Preprocessing Done Once Before Training	8
6.2.2	Label Convention Mismatch	9
6.2.3	Anchored Cropping	9
6.2.4	Per-Label Binary Warping	9
6.2.5	B-Spline Overshoot Clipping	9
6.2.6	Smoothness Weight λ	10
6.2.7	Training Schedule	10
6.3	CMRxM22	10
6.3.1	Image Quality Filtering	10
6.3.2	Label Remapping	10
6.3.3	Per-Label Binary Warping (Carried Over)	10
7	Evaluation Metrics	11
7.1	Dice Similarity Coefficient (DSC)	11
7.2	Average Symmetric Surface Distance (ASSD)	11
7.3	Hausdorff Distance (HD95)	11
7.4	Non-Positive Jacobian Determinant ($ J_\phi \leq 0$)	11
8	Results	12
8.1	Results on ACDC17	12
8.2	Results on M&Ms20	13
8.3	Results on CMRxM22	14
8.4	Ablation: Smoothness Weight on M&Ms20	15
8.5	Inference Speed	15
8.6	Qualitative Results	16
9	Discussion	16
9.1	Agreement with the Paper	16
9.2	Sources of Discrepancy	16
9.3	Hausdorff Distance Interpretation	16
9.4	Baseline Methods	16
10	Conclusion	16
A	Main Instructions to the AI Assistant	17

1 Introduction

1.1 What This Project Is About

This project takes a published cardiac registration method (DIR-MRVIT) and reproduces its results on three different MR datasets. The model code is from the paper’s GitHub repository. The rest of the pipeline (preprocessing, training driver, evaluation scripts) is written from scratch for each dataset.

1.2 Clinical Motivation

Cardiac MR imaging is the main way to look at the heart and to get diagnostic and prognostic information for heart disease. The motion of the heart muscle (the myocardium) during contraction and relaxation can be summarised with parameters such as strain and ejection fraction. Measuring this motion by hand is slow and varies between observers, so automatic methods are useful.

Estimating cardiac motion is hard for two reasons. First, the intensity inside the left ventricle is not uniform between time phases. Second, slice misalignments during acquisition produce large deformations inside one subject.

1.3 What Image Registration Does

Image registration finds a spatial mapping from one image (the *moving* image) to another (the *fixed* image). For cardiac motion estimation, the moving image is the end-systolic (ES) frame and the fixed image is the end-diastolic (ED) frame. The mapping produced by registration tells us how every voxel of the heart moved between the two phases.

Classical registration solves an optimisation problem for every image pair, which is slow. Learning-based registration trains a network once. After training, the network can register any new pair in a single forward pass.

DIR-MRVIT is a learning-based method. It combines a CNN with a Transformer in a three-level multi-resolution pyramid. The Transformer part captures long-range spatial structure. The pyramid lets the network refine the deformation from coarse to fine.

1.4 Project Objective

The task was to take the published DIR-MRVIT code, read the paper, and apply the method to three public CMR datasets: ACDC17, M&Ms20, and CMRxM22. The original repository only targets M&Ms20. The goal here is a direct comparison between the numbers obtained on all three datasets and the numbers reported in Tables 2–4 of the paper.

1.5 Use of AI Assistance

An AI assistant was used during implementation for translating design decisions into code and for resolving syntactic issues. The methodological choices — the split design, the training schedule, the smoothness weight, the interpretation of the Hausdorff distance, the cropping strategy — were taken by the author after reading the paper. Appendix A lists the main instructions given to the assistant.

2 Overview of the DIR-MRVIT Method

This section explains how the DIR-MRVIT method works at a high level, then gives the model details and the loss function. Figure 1 shows the overall data flow.

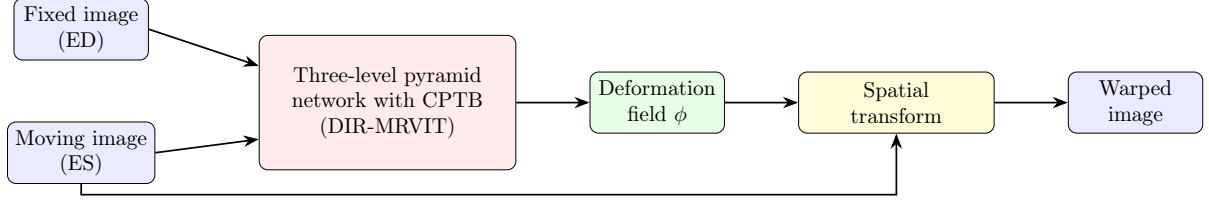


Figure 1: High-level data flow of DIR-MRVIT at inference time. The fixed image (ED) and the moving image (ES) are fed into the three-level pyramid network, which outputs a deformation field ϕ . The spatial transform applies ϕ to the moving image to produce the warped image, which should look like the fixed image. During training, a loss compares the warped image with the fixed image (see Section 2.5).

2.1 Problem Formulation

Let I_F and I_M be the fixed and moving images. Deformable registration looks for a deformation field ϕ that maps every voxel of I_M to a new location in I_F . The network is trained so that the warped moving image $I_M \circ \phi$ matches the fixed image:

$$\hat{\phi} = \arg \min_{\phi} E(I_F, I_M \circ \phi). \quad (1)$$

2.2 Three-Level Pyramid

DIR-MRVIT works at three resolutions, from coarse to fine:

- **Level 1 (coarsest):** $24 \times 24 \times 4$ voxels. The network sees the heavily downsampled images and predicts a rough deformation field ϕ_1 . This level captures the large overall motion.
- **Level 2 (medium):** $48 \times 48 \times 8$ voxels. The previous field ϕ_1 is upsampled and used as a starting guess. The network refines it into ϕ_2 .
- **Level 3 (finest):** $96 \times 96 \times 16$ voxels. Same idea: start from the upsampled ϕ_2 and refine to the final ϕ_3 .

2.3 Global–Local Cooperative Learning

At levels 2 and 3, two networks run in parallel. A main network looks at the whole image and predicts a global field. A sub-network looks at small patches and predicts a local field. The two are added together with the upsampled field from the previous level:

$$\phi_i = \tilde{\phi}_{i-1} + \phi'_i + \phi''_i. \quad (2)$$

The idea: the global path handles smooth, large-scale motion; the patch path handles local detail.

2.4 Convolutional Projection Transformer Block (CPTB)

The bottleneck of each U-shaped sub-network is a CPTB. A standard vision Transformer splits the input into patches, embeds them linearly, then runs self-attention. The CPTB replaces these linear embeddings with convolutions, and replaces the multi-layer perceptron with two more convolutions. This keeps the global modelling power of attention while staying convolutional.

For an input feature map X , three convolutions produce the query, key, and value:

$$Q = W^Q \cdot X, \quad K = W^K \cdot X, \quad V = W^V \cdot X. \quad (3)$$

Multi-head attention then computes

$$\text{SA}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right) V. \quad (4)$$

The result is added to the input, layer-normalised, and passed through two more convolutions:

$$Y = \text{Conv}(\text{Conv}(\text{LN}(\text{SA} + X))). \quad (5)$$

2.5 Loss Function

The total loss combines a similarity term and a regularisation term:

$$\mathcal{L}(f, m, \phi) = \mathcal{L}_{sim}(f, m \circ \phi) + \lambda \cdot \mathcal{R}(\phi). \quad (6)$$

The similarity term is a multi-resolution local normalised cross-correlation:

$$\mathcal{L}_{sim}(f, m \circ \phi) = - \sum_{i=1}^{\mathcal{N}} \frac{\text{LNCC}_\omega(f_i, m_i \circ \phi)}{2^{\mathcal{N}-i}}, \quad (7)$$

with window size $\omega = 2i + 1$ at level i . LNCC is robust to intensity inhomogeneity, which is important for cardiac MR.

The regulariser penalises sharp changes in the displacement field u :

$$\mathcal{R}(\phi) = \sum_{p \in \Omega} \|\nabla u(p)\|^2. \quad (8)$$

A large λ keeps the field smooth but limits the deformation; a small λ gives more freedom but allows folding.

2.6 Training Scheme

The pyramid is trained level by level. First the coarsest level is trained on its own. Then level 2 is added, and the level-1 weights are frozen for a number of iterations so the new level can stabilise. After the freeze ends, all weights are updated together. The same pattern is repeated when level 3 is added.

3 Datasets

This section lists the three datasets, then summarises them in Table 1. All registrations are between the end-diastolic (ED) and end-systolic (ES) phases. The ED image is the fixed image; the ES image is the moving image. Manual segmentation labels for the left-ventricle blood pool (LV) and the left-ventricle myocardium (MYO) are used as ground truth at evaluation time.

ACDC17: 100 short-axis cine MR images from the ACDC17 training set. Five pathology groups of 20 subjects each (healthy, infarction, dilated cardiomyopathy, hypertrophic cardiomyopathy, abnormal right ventricle).

M&Ms20: 150 short-axis cine MR images from the M&Ms20 challenge. Acquired on 1.5T and 3.0T scanners from six clinical centres. Covers a range of cardiovascular diseases and healthy volunteers. This is the most heterogeneous dataset.

	ACDC17	M&Ms20	CMRxM22
Subjects / pairs	100	150	67–69
In-plane resolution (mm)	0.70–1.92	0.85–1.45	0.66–0.76
Resampled spacing (mm)	$1.25 \times 1.25 \times 5$	$1.25 \times 1.25 \times 8$	$1.25 \times 1.25 \times 8$
Crop size (voxels)	$96 \times 96 \times 16$	$96 \times 96 \times 16$	$96 \times 96 \times 16$
Pathology groups	5	7	breath-hold conditions
Centres / scanners	1.5T & 3.0T	6 centres	single 3.0T

Table 1: Summary of the three CMR datasets used in this work. M&Ms20 includes six disease groups plus healthy volunteers (7 groups in total). For CMRxM22, 20 volunteers contribute 80 ED/ES pairs in total; 67–69 are kept after image quality filtering.

CMRxM22: From the CMRxMotion challenge; 20 healthy volunteers scanned under different breath-hold behaviours, so the images contain respiratory-motion artefacts. CMRxM22 is used only for testing: an M&Ms-trained model is applied directly to the CMRxM22 pairs, with no further training.

4 Pipeline Overview

Before going into the dataset details, Figure 2 shows the overall pipeline for all three datasets. ACDC17 and M&Ms20 are trained with five-fold cross-validation. CMRxM22 is not trained at all; the M&Ms20-trained model is applied directly to the CMRxM22 pairs.

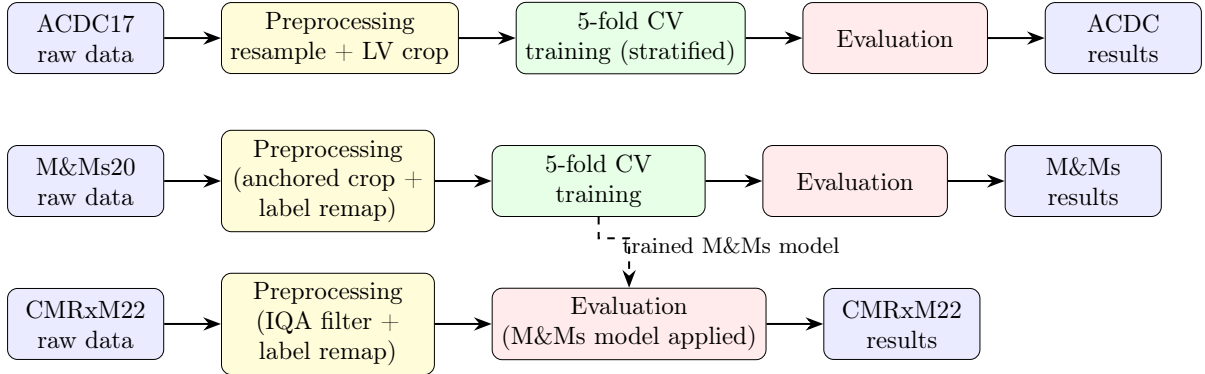


Figure 2: Pipeline overview for the three datasets. ACDC17 and M&Ms20 are trained with five-fold cross-validation. For CMRxM22, an M&Ms-trained fold model is loaded and applied directly to the ED/ES pairs (dashed arrow); there is no training on CMRxM22 data. Yellow blocks are preprocessing, green blocks are training, red blocks are evaluation.

5 Implementation Details

This section covers the parts that are the same across all three datasets: which code we reused, which code we wrote, the training setup, the common preprocessing steps, and the choice of Hausdorff distance. Dataset-specific changes are in Section 6.

5.1 Reused vs. Re-implemented Code

The model definition is used exactly as in the original repository:

- `GP_TF.py`: the three-level pyramid, the global main network, the local sub-network, the CPTB bottleneck.
- `TransBlock.py`: the CPTB.
- `Functions.py`: loss functions (multi-resolution LNCC, smoothness, negative Jacobian) and spatial transform.

These files were not modified.

The following scripts were written for this project:

- `cross_validation_acdc.py`: ACDC five-fold training driver with stratified split, on-the-fly resampling and cropping, and per-fold checkpoint saving.
- `cross_validation_mms.py`: M&Ms five-fold training driver.
- `preprocess_mms.py`: one-off M&Ms preprocessing (resampling, anchored cropping, label handling, intensity normalisation).
- `preprocess_cmrxm22.py`: CMRxM22 preprocessing with image quality filtering and label remapping to the M&Ms convention.
- `acdc_results.py`: ACDC evaluation on the sample data. Each patient is evaluated with the fold model whose test set it belongs to. It prints a per-patient table, a sample / full / paper comparison, and draws a qualitative figure for one patient.
- `mms_results.py`: M&Ms evaluation on the sample data, with the same table and figure output.
- `cmr_results.py`: CMRxM22 evaluation on the sample data. The M&Ms-trained model is applied directly; by default all five fold models are averaged.

5.2 Training Setup

Training uses the Adam optimiser with learning rate 10^{-4} and batch size one (as in the paper). The pyramid is trained level by level: when a new level is added, the lower level is frozen for a number of iterations, then unfrozen. The freeze step is set to 5000 iterations for ACDC and 8000 for M&Ms (the more heterogeneous data takes longer to stabilise). Validation Dice is monitored, and the best checkpoint per fold is kept. All experiments ran on a single NVIDIA RTX 5070 Ti GPU.

5.3 Common Preprocessing

For all three datasets, images are resampled to the spacings in Table 1 and cropped to $96 \times 96 \times 16$ around the LV. Intensities are scaled to $[0, 1]$. The dataset-specific differences (anchored cropping for M&Ms, label remapping for CMRxM22, and so on) are explained in Section 6.

5.4 Hausdorff Distance: HD95 vs Strict Maximum

The paper defines HD as the strict maximum in Equation 10. In practice, the strict maximum is dominated by a single outlier voxel. With 8 mm inter-slice spacing in M&Ms20 and CMRxM22, the strict maximum is much larger than the values shown in the paper’s tables (typically 5–8 mm). These reported magnitudes are consistent with HD95, not with the strict maximum.

HD95 is also the standard convention in the medical image registration literature. For these two reasons, HD95 is used throughout this work, and this decision is stated clearly.

6 Dataset-Specific Modifications

The paper provides the model and a small demo dataset. The rest (preprocessing for each dataset, the cross-validation, and the evaluation) had to be built. Each dataset also has its own quirks. This section lists what was changed or added for ACDC17, M&Ms20, and CMRxM22.

6.1 ACDC17

6.1.1 Resampling and Cropping

Each frame is resampled to $1.25 \times 1.25 \times 5$ mm (B-spline for the image, nearest-neighbour for the segmentation). A $96 \times 96 \times 16$ box around the LV centre is cropped. Intensities are then scaled to $[0, 1]$.

6.1.2 Stratified Five-Fold Split

ACDC17 stores its 100 subjects in pathology blocks of 20:

- subjects 0–19: healthy (NOR)
- subjects 20–39: infarction (MINF)
- subjects 40–59: dilated cardiomyopathy (DCM)
- subjects 60–79: hypertrophic cardiomyopathy (HCM)
- subjects 80–99: abnormal right ventricle (ARV)

A simple sequential split puts each fold’s test set entirely inside one pathology. This means each fold is trained on four pathologies and tested on a fifth. Early runs with this split gave poor results; some folds barely converged.

The fix is a stratified split: each fold’s test set has four subjects from each pathology, so every fold has the same pathology balance. A fixed random seed is used inside each block, so the split is reproducible.

6.1.3 Training Schedule

The freeze step is 5000 iterations per level. Early stopping with a patience of 3000 iterations is added to the level-3 stage to avoid overfitting and to save time.

6.1.4 Surface Metrics

The paper splits the surface metrics into the LV endocardium (the inner LV boundary) and the LV epicardium (the outer myocardial boundary). For ACDC17 the labels are $LV = 3$ and $MYO = 2$:

- Endocardium: surface of label 3 (LV alone).
- Epicardium: surface of $(\text{label } 2 \cup \text{label } 3)$ (LV + myocardium).

The same convention is used in this work.

6.2 M&Ms20

6.2.1 Preprocessing Done Once Before Training

M&Ms20 stores each subject as one 4D file with a time axis (20–30 frames per subject). The ED and ES frame indices are listed in a CSV. Preprocessing each pair on every training iteration would be very slow.

Instead, all preprocessing is done once before training: read the CSV, extract the ED and ES frames, resample to $1.25 \times 1.25 \times 8$ mm (note the thicker slices, 8 mm not 5 mm), crop, normalise, and save four files per subject (`_ED.nii.gz`, `_ED_gt.nii.gz`, `_ES.nii.gz`, `_ES_gt.nii.gz`). Training reads these saved files directly.

6.2.2 Label Convention Mismatch

M&Ms20 uses different label numbers than ACDC17:

Structure	ACDC17	M&Ms20
LV blood pool	3	1
RV blood pool	1	2
LV myocardium	2	3

The ACDC pipeline had “LV = label 3” hard-coded in the crop function and the Dice calculation. When this code was first run on M&Ms20, the crop was centred on the myocardium (M&Ms label 3) instead of the LV (M&Ms label 1), and the LV Dice was actually measuring the MYO. The first run gave LV and MYO Dice around 0.61.

After the labels were fixed, the LV Dice jumped to 0.74. This is a small but very common kind of bug when moving code between datasets. The wrong number (0.61) was not very low, so it looked at first like a model that simply did not train well.

6.2.3 Anchored Cropping

In the ACDC pipeline, the ED and ES frames were cropped independently. Each frame’s crop box was computed from its own LV mask. For M&Ms this turned out to be wrong: the LV centre shifts between ED and ES, so independent cropping silently removes part of the real motion the model has to learn.

For M&Ms the crop box is now computed from the ED frame only, and the same box is used for the ES frame. This keeps the spatial relation between ED and ES intact. Intensity normalisation is also moved to after cropping, so the $[0, 1]$ range covers only the cardiac region (and not bright artefacts in the full field of view).

6.2.4 Per-Label Binary Warping

The first M&Ms evaluation gave Dice around 0.74 but HD95 around 44 mm. That HD is much too large for these structures. The cause was the segmentation warping.

The original code warped the multi-label segmentation as one image with bilinear interpolation, then rounded to the nearest integer label. The problem: an LV voxel (label 1) next to a MYO voxel (label 3), after bilinear interpolation and rounding, can produce a value of 2 (the RV). So the warped segmentation ended up with false RV voxels along the LV/MYO border. These false voxels do not exist in any real anatomy, but they dominate HD95.

The fix is per-label binary warping. Each label is warped separately as a binary mask. The warped masks are then combined by taking the label with the highest warped probability at each voxel. No intermediate label values are produced. After this fix, HD95 dropped from about 44 mm to about 7 mm.

6.2.5 B-Spline Overshoot Clipping

B-spline resampling can produce small negative values around bright edges. These negatives confuse the $[0, 1]$ normalisation step. The resampled images are clipped to be non-negative before normalisation.

6.2.6 Smoothness Weight λ

The paper uses $\lambda = 0.5$, tuned on ACDC17. For M&Ms20, $\lambda = 0.5$ turned out to over-regularise the deformation: the field could not produce strong enough deformations. Two reasons:

- M&Ms20 has thicker slices (8 mm vs 5 mm), so each voxel covers more tissue and the deformation has to be larger to move the same anatomy.
- M&Ms20 is multi-centre and multi-disease, so the deformation patterns vary more across subjects.

A lower $\lambda = 0.2$ gives the model more freedom to produce strong local deformations. The choice is supported by an ablation (Section 8.4).

6.2.7 Training Schedule

The freeze step is raised from 5000 (ACDC) to 8000 for M&Ms20: the more complex data takes longer to converge at each level. Early stopping uses a patience of 5000 iterations on the validation Dice.

6.3 CMRxM22

CMRxM22 is not used for training. Following the paper, a model trained on M&Ms20 is loaded and applied directly to the CMRxM22 pairs, with no fine-tuning. We report the average over all five M&Ms-trained fold models for a more stable estimate, instead of the single randomly chosen fold model used in the paper.

6.3.1 Image Quality Filtering

CMRxM22 deliberately contains scans with strong respiratory-motion artefacts. Some images are not usable for clinical assessment. The challenge provides an `IQA.csv` file that rates each image from 1 (excellent) to 4 (non-diagnostic).

Score-4 cases were excluded during preprocessing. Score-3 cases were kept, so the artefact-heavy regime is still represented.

6.3.2 Label Remapping

CMRxM22 uses yet another label convention:

Structure	ACDC17	M&Ms20	CMRxM22
LV blood pool	3	1	1
RV blood pool	1	2	3
LV myocardium	2	3	2

The model was trained on M&Ms20 with MYO = label 3. If the CMRxM22 ground truth is read as is, the evaluation compares the model’s predicted MYO (label 3) against the CMRxM22 RV (also label 3). This gave an artificially high MYO Dice of 0.758 in early runs, because the RV and the MYO overlap to some extent.

To get an honest comparison, the CMRxM22 ground-truth labels are remapped to the M&Ms20 convention during preprocessing: labels 2 and 3 are swapped (LV stays at 1). After this remap, the MYO Dice dropped to 0.641 — lower, but now measuring the correct structure.

6.3.3 Per-Label Binary Warping (Carried Over)

The same per-label binary warping used for M&Ms20 is used here too. This is one of the few things the M&Ms model brings with it to CMRxM22 for free.

7 Evaluation Metrics

Four metrics are used. They match the paper. All metrics are computed by applying the predicted deformation field to the moving image’s manual segmentation, then comparing the result against the fixed image’s ground-truth segmentation.

7.1 Dice Similarity Coefficient (DSC)

DSC measures volume overlap between two binary segmentations:

$$\text{DSC}(V_P, V_G) = \frac{2|V_P \cap V_G|}{|V_P| + |V_G|}. \quad (9)$$

DSC is between 0 and 1. Higher is better. Two structures are scored separately: the LV blood pool and the LV myocardium. MYO is thinner and harder to register, so MYO Dice is always lower than LV Dice.

7.2 Average Symmetric Surface Distance (ASSD)

DSC can miss small but spatially large boundary errors. ASSD measures the average distance between two surfaces, in both directions:

$$\text{ASSD}(S_P, S_G) = \frac{\sum_{p \in S_P} d(p, S_G) + \sum_{g \in S_G} d(g, S_P)}{|S_P| + |S_G|}, \quad (10)$$

where $d(p, S_G) = \min_{g \in S_G} \|p - g\|_2$. The result is in millimetres, using the physical voxel spacing. Lower is better.

Two contours are evaluated: the LV endocardium (inner LV boundary) and the LV epicardium (outer myocardial boundary, surface of $\text{LV} \cup \text{MYO}$).

7.3 Hausdorff Distance (HD95)

HD captures the worst-case boundary error. The strict maximum is

$$\text{HD}(S_P, S_G) = \max\left\{\max_{p \in S_P} d(p, S_G), \max_{g \in S_G} d(g, S_P)\right\}, \quad (11)$$

which is set by the single most distant surface point pair and is very sensitive to outliers. The 95th-percentile version drops the most extreme 5% of distances:

$$\text{HD95}(S_P, S_G) = \text{percentile}_{95}\left(\{d(p, S_G) : p \in S_P\} \cup \{d(g, S_P) : g \in S_G\}\right). \quad (12)$$

HD95 is used throughout, for the reasons given in Section 5. The result is in millimetres. Lower is better.

7.4 Non-Positive Jacobian Determinant ($|J_\phi| \leq 0$)

The Jacobian determinant of the deformation field at each voxel measures local volume change. For a smooth, invertible deformation, every Jacobian determinant should be positive. A non-positive value means the deformation has folded at that voxel, which is not physically valid.

With $\phi = \text{id} + u$, the Jacobian at voxel p is

$$J_\phi(p) = I + \nabla u(p), \quad (13)$$

where $\nabla u(p)$ is the 3×3 gradient matrix of the displacement. The reported metric is the percentage of voxels with $\det(J_\phi) \leq 0$:

$$|J_\phi| \leq 0 (\%) = \frac{100}{|\Omega|} \sum_{p \in \Omega} \mathbf{1}[\det(J_\phi(p)) \leq 0]. \quad (14)$$

Lower is better. This metric reflects a trade-off: a very smooth field has low $|J_\phi| \leq 0$ but may miss complex motion, while a very aggressive field improves overlap but folds more.

8 Results

Each dataset is evaluated with a dedicated results script. Alongside the full-dataset numbers reproduced in this project, the same script also runs on the small sample subset shipped with the repository (10 ACDC, 10 M&Ms, 18 CMR_xM22) and prints a sample / full / paper comparison. The sample column is a quick sanity check that the released code and checkpoints reproduce the same behaviour; the full column is the actual reproduction result. HD is reported as HD95 throughout.

8.1 Results on ACDC17

Table 2 compares our results against the paper on the 100 ACDC17 pairs, together with the sample-data result. LV and MYO Dice are slightly below the paper. HD95 is better than the paper. The non-positive Jacobian percentage is lower, so the deformation field is smoother. The sample column is close to the full result; the LV Dice is a little lower because the 10 sample patients are dominated by the harder infarction group.

Metric	Sample (10)	Full (100)	Paper (Proposed)
LV Dice	0.865 ± 0.084	0.887 ± 0.069	0.917 ± 0.043
MYO Dice	0.756 ± 0.062	0.752 ± 0.065	0.789 ± 0.055
Endo ASSD (mm)	2.02 ± 1.23	1.51 ± 1.03	0.79 ± 0.39
Epi ASSD (mm)	1.58 ± 0.67	1.08 ± 0.51	0.88 ± 0.21
Endo HD (mm)	6.19 ± 3.00	5.27 ± 2.43	5.62 ± 1.22
Epi HD (mm)	5.70 ± 2.44	4.40 ± 2.01	5.51 ± 1.77
$ J_\phi \leq 0 (\%)$	0.14 ± 0.17	0.13 ± 0.16	0.41 ± 0.15

Table 2: ACDC17 results (100 pairs, five-fold cross-validation). The sample column is the 10-patient sample subset. HD reported as HD95.

The qualitative result for one ACDC patient is shown in Figure 3.

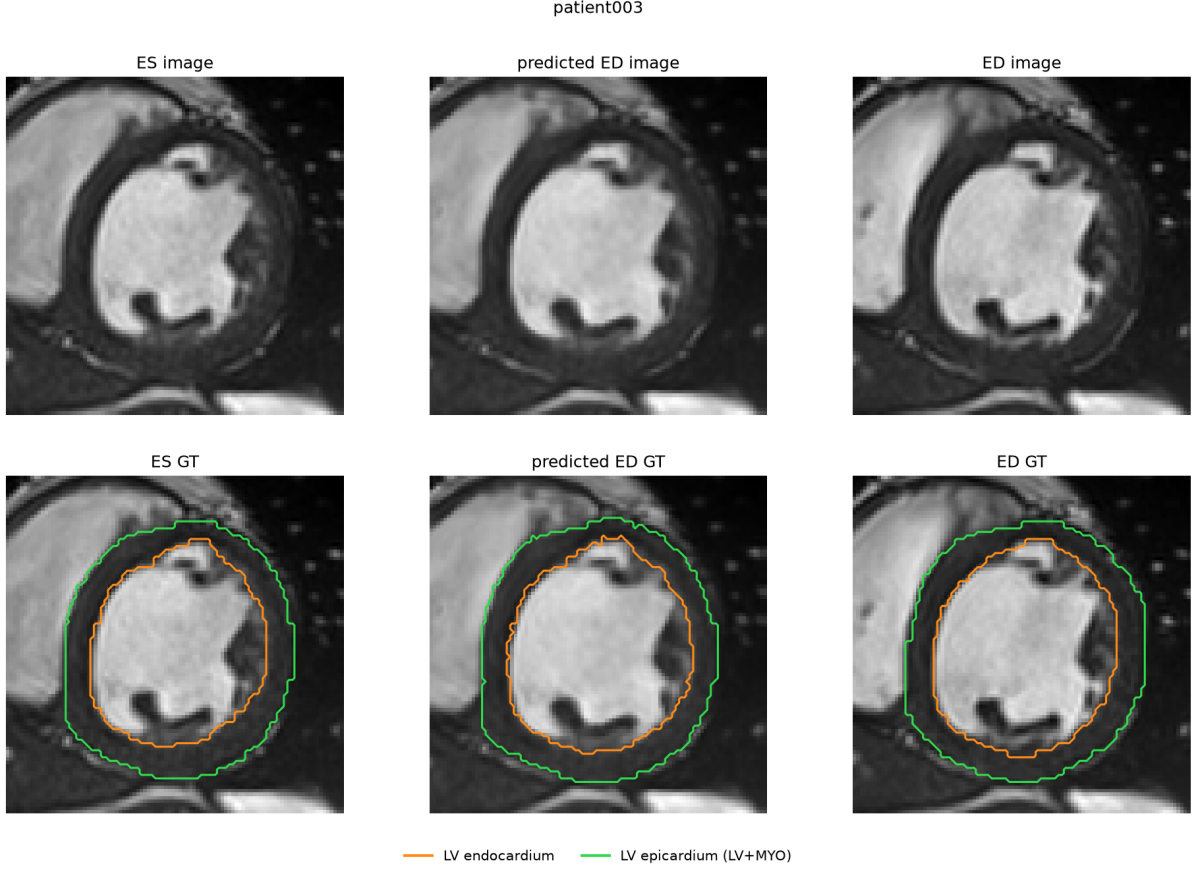


Figure 3: ACDC17 qualitative result for one patient. Top row: ES image, predicted ED image (warped ES), and ED image. Bottom row: the same three frames with the LV endocardium (orange) and LV epicardium (green) contours. The predicted ED contours are obtained by warping the ES segmentation with the estimated field; they should match the ED ground-truth contours.

8.2 Results on M&Ms20

Table 3 shows the M&Ms20 results. The reproduction is very close to the paper: LV Dice is within 0.007. The endocardium surface metrics are slightly better than the paper, and the epicardium metrics are slightly worse. The sample column agrees with the full result.

Metric	Sample (10)	Full (150)	Paper (Proposed)
LV Dice	0.891 ± 0.026	0.877 ± 0.050	0.884 ± 0.038
MYO Dice	0.780 ± 0.054	0.790 ± 0.070	0.729 ± 0.057
Endo ASSD (mm)	1.29 ± 0.30	1.62 ± 0.92	1.88 ± 0.79
Epi ASSD (mm)	1.93 ± 0.49	2.08 ± 0.74	2.05 ± 0.83
Endo HD (mm)	6.86 ± 1.41	7.15 ± 2.77	7.40 ± 2.78
Epi HD (mm)	8.13 ± 0.84	8.74 ± 2.28	8.16 ± 1.95
$ J_\phi \leq 0$ (%)	0.24 ± 0.15	0.28 ± 0.20	0.54 ± 0.32

Table 3: M&Ms20 results (150 pairs, five-fold cross-validation, $\lambda = 0.2$). The sample column is the 10-patient sample subset. HD reported as HD95.

The qualitative result for one M&Ms patient is shown in Figure 4.

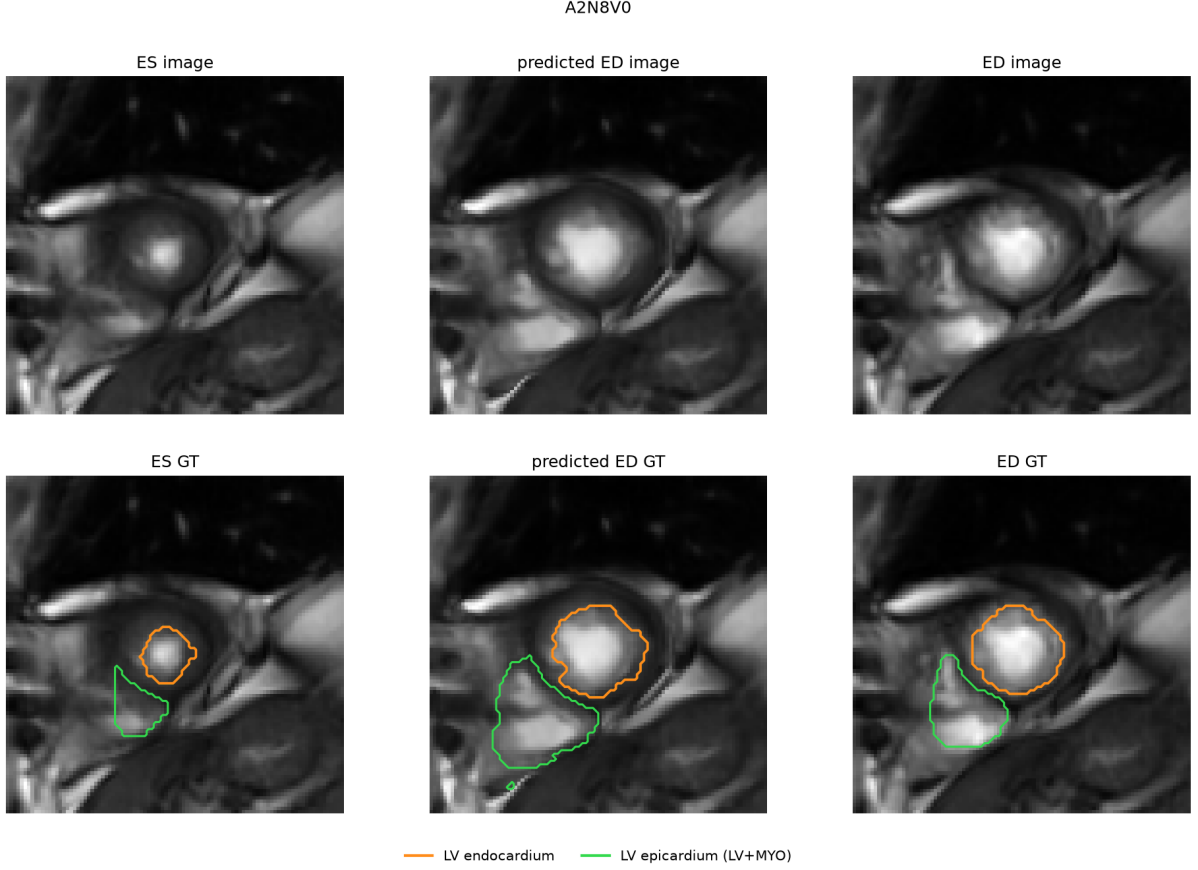


Figure 4: M&Ms20 qualitative result for one patient, in the same layout as Figure 3.

8.3 Results on CMRxM22

Following the paper, CMRxM22 is evaluated with a model trained on M&Ms20, with no fine-tuning. The average over all five M&Ms-trained fold models is reported in Table 4. LV Dice and epicardium surface metrics are slightly behind the paper.

Metric	Sample (18)	Full	Paper
LV Dice	0.871 ± 0.035	0.857 ± 0.051	0.892 ± 0.027
MYO Dice	0.825 ± 0.031	0.758 ± 0.077	0.703 ± 0.050
Endo ASSD (mm)	1.86 ± 0.72	1.95 ± 0.86	1.78 ± 0.60
Epi ASSD (mm)	2.12 ± 0.60	2.63 ± 0.82	1.94 ± 0.69
Endo HD (mm)	7.19 ± 1.98	7.51 ± 2.11	7.82 ± 2.31
Epi HD (mm)	7.79 ± 1.06	9.41 ± 2.63	8.07 ± 2.24
$ J_\phi \leq 0$ (%)	0.90 ± 0.44	1.05 ± 0.51	0.30 ± 0.14

Table 4: CMRxM22 results. The M&Ms-trained model is applied directly to the CMRxM22 pairs. The full column is the average across all five M&Ms-trained fold models. HD reported as HD95.

The qualitative result for one CMRxM22 case is shown in Figure 5.

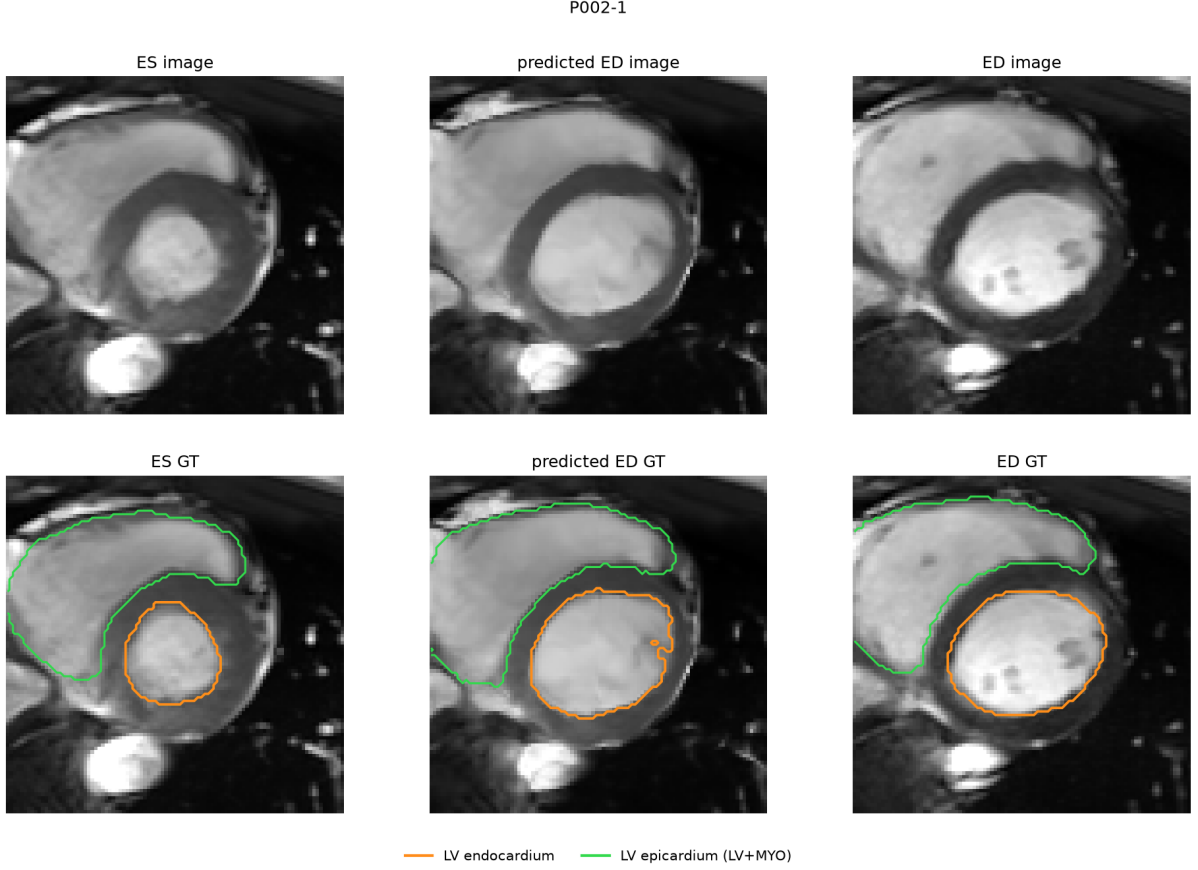


Figure 5: CMRxM22 qualitative result for one case, in the same layout as Figure 3.

8.4 Ablation: Smoothness Weight on M&Ms20

Table 5 compares $\lambda = 0.5$ (the paper’s value) with $\lambda = 0.2$ on M&Ms20, both with five-fold cross-validation and HD95. Lowering λ improves every overlap and surface metric. The lower smoothness weight is better suited to the thicker and more heterogeneous M&Ms20 data.

Metric	$\lambda = 0.5$	$\lambda = 0.2$	Paper
LV Dice	0.845 ± 0.059	0.877 ± 0.050	0.884 ± 0.038
MYO Dice	0.766 ± 0.076	0.790 ± 0.070	0.729 ± 0.057
Endo ASSD (mm)	2.07 ± 0.96	1.62 ± 0.92	1.88 ± 0.79
Epi ASSD (mm)	2.45 ± 0.82	2.08 ± 0.74	2.05 ± 0.83
Endo HD (mm)	8.08 ± 2.33	7.15 ± 2.77	7.40 ± 2.78
$ J_\phi \leq 0$ (%)	0.04	0.28 ± 0.20	0.54 ± 0.32

Table 5: Effect of the smoothness weight λ on M&Ms20 (five-fold, HD95).

8.5 Inference Speed

Using the original test routine adapted to the project’s data format, the average registration time is about 0.086s per image pair on the RTX 5070 Ti. This is very close to the 0.092s per pair reported in the paper.

8.6 Qualitative Results

Figures 3, 4, and 5 show one example case per dataset. Each figure has two rows. The top row shows the ES image, the predicted ED image (the ES image warped by the estimated field), and the ED image. The bottom row shows the same three frames overlaid with two contours: the LV endocardium (orange) and the LV epicardium (green). For the predicted ED column, the contours come from warping the ES segmentation; for the ES and ED columns they are ground truth. A good registration makes the predicted ED contours match the ED ground-truth contours.

9 Discussion

9.1 Agreement with the Paper

The reproduction works across all three datasets. The M&Ms20 results are the closest to the paper: LV Dice is within 0.007. HD95 is competitive throughout.

9.2 Sources of Discrepancy

Several things explain the small remaining differences:

- **Unspecified training schedule:** the paper does not say exactly how many iterations the proposed model was trained for. This work used a fixed iteration budget with early stopping on validation Dice. The model may not have reached the same point as in the original work.
- **Random training variation:** random pair sampling, weight initialisation, and batch ordering all add a ± 0.01 – 0.02 Dice spread from run to run.
- **ACDC cropping:** for ACDC17 the ED and ES frames were cropped independently, while the anchored cropping in M&Ms20 keeps inter-frame displacement intact. This is consistent with ACDC17 showing the largest Dice gap to the paper. Applying anchored cropping to ACDC17 as well would be a clear next step.
- **Channel configuration:** the exact link between the paper’s channel parameter and the repository’s `start_channel` is not fully specified, which may affect model capacity.

9.3 Hausdorff Distance Interpretation

The choice to report HD95 instead of the strict maximum is the single most important metric decision. The strict maximum is not consistent with the magnitudes the paper reports, and HD95 is the standard convention in the field. The strict maximum would have given HD values roughly twice as large for all methods, which does not match the paper’s tables.

9.4 Baseline Methods

The paper compares the proposed method against seven baselines (SyN, NiftyReg, Elastix, PTVReg, VoxelMorph, LapIRN, TransMorph). These baselines were not re-implemented here. The aim was to reproduce the proposed method, and the published baseline numbers serve as the comparison reference.

10 Conclusion

This project reproduced the DIR-MRVIT deformable registration method and evaluated it on three CMR datasets following the paper’s protocols. The original model code was used

unchanged. The preprocessing, the cross-validation, and the evaluation pipelines were re-implemented and adapted to each dataset. The reproduced results are close to the paper. Differences are small and come from the unspecified training schedule, random variation, and minor preprocessing choices. An ablation on the smoothness weight justifies $\lambda = 0.2$ for the heterogeneous M&Ms20 data. Possible improvements that stay faithful to the paper include applying anchored cropping to ACDC17 and adding test-time augmentation, both of which work at inference time and need no retraining.

A Main Instructions to the AI Assistant

The list below records the main technical instructions given to the AI assistant during implementation. Each instruction reflects a methodological decision taken by the author. The assistant was used to translate these decisions into code.

Instruction: ACDC has five pathology groups of twenty subjects each. Make the five-fold split stratified so every fold’s test set has four subjects from each group, and fix the random seed so the split is reproducible.

Instruction: M&Ms is 4D with per-subject ED/ES frame indices in the CSV. Move preprocessing out of the training loop: read the frame indices, resample each frame to $1.25 \times 1.25 \times 8$ mm, crop, normalise, and save. For the crop, compute the box from the ED label only and apply the same box to ES, so the inter-frame displacement is preserved.

Instruction: M&Ms uses a different label convention from ACDC (LV = 1, RV = 2, MYO = 3). Fix the labels in the cross-validation and evaluation code. Also use per-label binary warping during evaluation: warp each label’s binary mask separately and take the arg-max, so that interpolation at boundaries does not produce labels that don’t exist.

Instruction: The paper’s $\lambda = 0.5$ was tuned on ACDC. M&Ms has 8 mm slices and multi-centre deformations, so the field is likely over-regularised. Run M&Ms with $\lambda = 0.2$ as well and report both, so the choice can be justified as an ablation.

Instruction: CMRxM22 has a different label convention again (LV 1, MYO 2, RV 3) and is evaluated with the M&Ms-trained model. Remap the CMRxM22 labels to the M&Ms convention during preprocessing, and filter out the non-diagnostic-quality cases using the IQA file. Add an option to run all five fold models and average, since the paper used a single randomly selected fold.